Retinal Blood Vessel Segmentation

Improving the CAR-UNet Paper on the FIVES Dataset

Complete project explanation — what we did, how, and why

	Paper model (CAR-UNet)	Our improved model	Change
Dice (overlap with expert)	0.8382	0.8436	+0.54 points
Sensitivity (vessels found)	0.8310	0.8465	+1.55 points
clDice (vessel connectivity)	0.8626	0.8751	+1.25 points
Dice vs 1024×1024 ground truth (thin vessels)	0.8572	0.8796	+2.24 points

Base paper: Guo et al., Channel Attention Residual U-Net for Retinal Vessel Segmentation, IEEE ICASSP 2021 (arXiv:2004.03702)
Dataset: FIVES (Jin et al., Scientific Data, 2022) — 800 fundus images
Course: 22AIE437 Medical Image Processing

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1. The project in one page

The problem. Doctors look at photographs of the back of the eye (the retina, photographed as a fundus image) to detect diseases such as diabetic retinopathy, glaucoma, age-related macular degeneration and hypertension. A key clue is the network of tiny blood vessels: their width, branching and breaks change with disease. Tracing every vessel by hand takes an expert a long time. Our goal is a program that takes a fundus photograph and automatically produces a black-and-white **vessel map** (white = vessel, black = background). This task is called **segmentation**: deciding, for every pixel, whether it belongs to a vessel.

The approach. We use deep learning — a type of neural network called a **U-Net** and its improved version **CAR-UNet** — trained on photographs that experts have already traced. The network learns from these examples and can then trace new images on its own.

The three stages

Stage	What we did	Result
Review 1	Rebuilt the original U-Net (Ronneberger et al., 2015) and trained it on the small DRIVE dataset (40 images).	Working baseline (Dice 0.759 on 4 images)
Stage 2 (paper)	Rebuilt the CAR-UNet paper (U-Net + residual blocks + channel attention) on the much larger FIVES dataset (800 images), fixed the evaluation so the comparison is fair, and compared it with U-Net.	CAR-UNet Dice 0.8382 vs U-Net 0.8344
Stage 3 (improvements)	Found 5 weaknesses of the paper and fixed 4 of them with new techniques (dual attention, clDice loss, higher resolution, tuned threshold), plus a much stronger evaluation.	Improved model Dice 0.8436, connectivity +1.25, thin vessels +2.24

Why the improvements matter

- The improved model is better on **9 of 11** metrics, and every difference is **statistically significant**.
- Its biggest gains are on **thin vessels and vessel connectivity** — exactly where the paper model fails and where early disease signs appear.
- Every claim is backed by 3 independent training runs (seeds), 200 test images and paired significance tests.

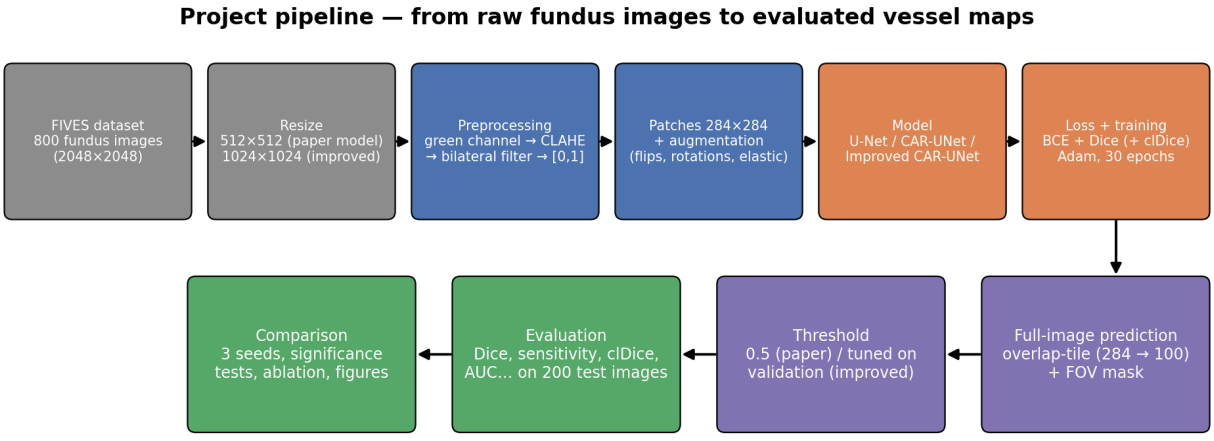


Figure 1. The whole pipeline from raw images to evaluated results. Each box is explained in a later section.

2. Background you need (explained simply)

2.1 Images as numbers

A digital image is a grid of numbers. A colour fundus image has three channels (red, green, blue); each pixel holds three values from 0 to 255. A vessel map is one channel with values 0 (background) or 1 (vessel). Segmentation means learning a function that turns the first grid into the second.

2.2 Convolutional neural networks (CNNs)

- **Convolution**: a small window (e.g. 3×3) of learned weights slides over the image and computes a weighted sum at each position. Early layers learn simple patterns (edges, dark lines); deeper layers combine them into complex patterns (branching vessels). The output of a layer is a **feature map**; a layer with 64 filters produces 64 feature maps (called **channels**).
- **ReLU**: sets negative values to zero. It adds non-linearity, so the network can learn more than straight-line relationships.
- **Max-pooling (2×2)**: keeps the largest value in each 2×2 block, halving the image size. This lets later layers see a larger region of the image.
- **Batch normalisation (BN)**: rescales each layer's outputs to a stable range. Training becomes faster and more stable.
- **Dropout**: randomly switches off some neurons during training, so the network cannot memorise the training images (reduces overfitting).
- **Sigmoid**: squashes the final output into 0–1, read as the probability that a pixel is a vessel.

2.3 How a network learns

- **Loss function**: a number measuring how wrong the prediction is compared with the expert tracing (lower = better).
- **Gradient descent / Adam**: the training algorithm repeatedly nudges every weight in the direction that reduces the loss. Adam is a popular variant that adapts the step size for each weight automatically.
- **Learning rate**: the size of each nudge. **Epoch**: one pass over the training data. **Batch**: the number of examples processed together.
- **Overfitting**: the network memorises training images instead of learning general rules; it then performs badly on new images. We fight it with augmentation, dropout and by choosing the best model on separate validation images.
- **Training / validation / test sets**: training images teach the network; validation images are used to make choices (best epoch, threshold); test images are used **only once at the end** to report honest results.

3. Stage 1 (Review 1): U-Net on DRIVE

In Review 1 we implemented the original **U-Net** (Ronneberger, Fischer and Brox, MICCAI 2015) and trained it on **DRIVE**, a classic benchmark of 40 retinal images (565×584 pixels). We used the same preprocessing and patch strategy described in Sections 5–7. The 20 DRIVE training images were split into 16 for training and 4 for validation; the model was trained for 5 epochs on a CPU (about 13 minutes).

Metric (4 DRIVE validation images)	Value
Accuracy	0.9399
Sensitivity	0.7761
Specificity	0.9628
Dice	0.7588
AUC-ROC	0.9565

Why we moved on from this stage
DRIVE is tiny (20 training images, one camera, mainly diabetic screening), only 4 images were used for evaluation, and only 5 CPU epochs were possible. Results on 4 images are not reliable, and a model trained on 16 images cannot generalise to different diseases. This motivated the larger FIVES dataset and a stronger model (CAR-UNet) in Stage 2.

4. Stage 2: the paper — CAR-UNet on FIVES

4.1 Why the FIVES dataset?

Dataset	Images	Resolution	Diseases	Why (not) used
DRIVE	40	565×584	mainly diabetic	Too small; used only in Review 1
STARE / CHASE_DB1	20 / 28	≈700–1000 px	few	Even smaller; not multi-disease
HRF	45	3504×2336	3 groups	High resolution but very few images
FIVES	**800**	**2048×2048**	**AMD, DR, Glaucoma, Normal**	**20× more images, 4 balanced disease groups, expert consensus labels → chosen**

FIVES (Jin et al., Scientific Data 2022) has 800 images from 573 patients, traced by a consensus of 3 ophthalmologists and 24 medical staff, with 200 images each of AMD, diabetic retinopathy (DR), glaucoma and normal eyes. More, and more varied, data is the single most reliable way to make a model generalise. Our sir confirmed that the project should use FIVES only.

4.2 Why CAR-UNet?

CAR-UNet (Guo et al., IEEE ICASSP 2021) is a direct, well-motivated upgrade of U-Net designed specifically for retinal vessels. It keeps U-Net's shape and adds two ideas: **residual blocks** (easier training of deep networks) and **channel attention** (the network learns which features are useful). It reported state-of-the-art results on DRIVE, CHASE_DB1 and STARE, and is simple enough to implement, understand and modify within a course project.

Alternative	Why not chosen as the base
Plain U-Net	Already done in Review 1; no attention, no residual learning
Attention U-Net / SA-UNet	Good options; CAR-UNet was the paper approved for the project
Transformer models (TransUNet, Swin-UNet)	Need far more data and GPU memory than a 6 GB laptop GPU; harder to explain and train
Classical filters (Frangi, matched filters)	Much lower accuracy than deep learning on modern benchmarks

5. Data preparation and preprocessing

5.1 Resizing

FIVES images are 2048×2048 pixels. Training on them directly is too slow and memory-hungry for a 6 GB laptop GPU, so we made smaller copies: **512×512** for the paper model and **1024×1024** for the improved model (Improvement 3). Images are shrunk with **INTER_AREA** interpolation (averages neighbouring pixels → smooth, no aliasing); masks are shrunk with **nearest-neighbour** interpolation so they stay strictly 0/1 (other methods would create in-between grey values that are neither vessel nor background).

5.2 Preprocessing steps (why each one)

Step	What it does	Why	Alternative and why not
1. Green channel	Keep only the green colour channel	Haemoglobin absorbs green light most strongly, so vessels have the highest contrast in green. Red is too bright/saturated; blue is dark and noisy.	Grey-scale or all 3 channels: lower vessel contrast, more noise
2. CLAHE (clip 2.0, 8×8 tiles)	Contrast Limited Adaptive Histogram Equalisation: boosts contrast locally in each tile	Fundus images are unevenly lit (bright centre, dark edges); local enhancement makes faint thin vessels visible everywhere. The clip limit stops noise from being amplified.	Global histogram equalisation: over-brightens the optic disc and amplifies noise
3. Bilateral filter (d=5, σ=75)	Smooths noise while keeping edges sharp	Averages only similar neighbouring pixels, so camera noise is removed but vessel borders are not blurred.	Gaussian blur: also blurs thin vessels away; median filter: can break 1-pixel vessels
4. Normalise to [0, 1]	Divide by 255	Neural networks train more stably with small input values	—
5. FOV mask	Mask of the circular retina region (threshold + morphology)	Pixels outside the circle are black and meaningless; metrics are computed only inside the field of view (FOV), otherwise accuracy would look artificially high.	—



Figure 2. Preprocessing of one FIVES image: RGB → green channel → CLAHE → bilateral filter → normalised, next to the expert ground truth.

5.3 Splitting the data (and avoiding data leakage)

- FIVES provides 600 training and 200 test images (50 per disease). The **test images are never used for training or for any decision**.
- The 600 training images are split into **480 for training and 120 for validation**, **stratified by disease** (30 validation images of each disease) with a fixed random seed, so every model sees exactly the same split.
- The split is made **per image, not per patch**. If patches of the same image were in both training and validation, the network would be validated on almost the same pixels it learned from (**data leakage**), giving falsely high scores.
- Why stratify? In the first version the validation set accidentally contained only Normal and AMD images, so the best model was chosen without ever checking DR or Glaucoma. Stratification guarantees all four diseases are represented.

6. Patches, augmentation and full-image prediction

6.1 Why train on patches?

Instead of whole images, the network is trained on small square **patches** cut at random positions inside the retina. Patches fit in GPU memory, give thousands of different training examples from 480 images, and vessels are a local pattern, so a patch contains all the information needed.

6.2 Valid convolutions: 284 → 100

We follow the original U-Net design with **valid convolutions** (no zero-padding). Each 3×3 convolution shrinks the image by 2 pixels; over the whole network the total shrinkage is 184 pixels, so a **284×284 input patch produces a 100×100 output** for its centre. Every output pixel is therefore predicted from 92 pixels of real image context on every side. With zero-padding, pixels near patch borders would be predicted from artificial zeros, which creates visible seams in the stitched result.

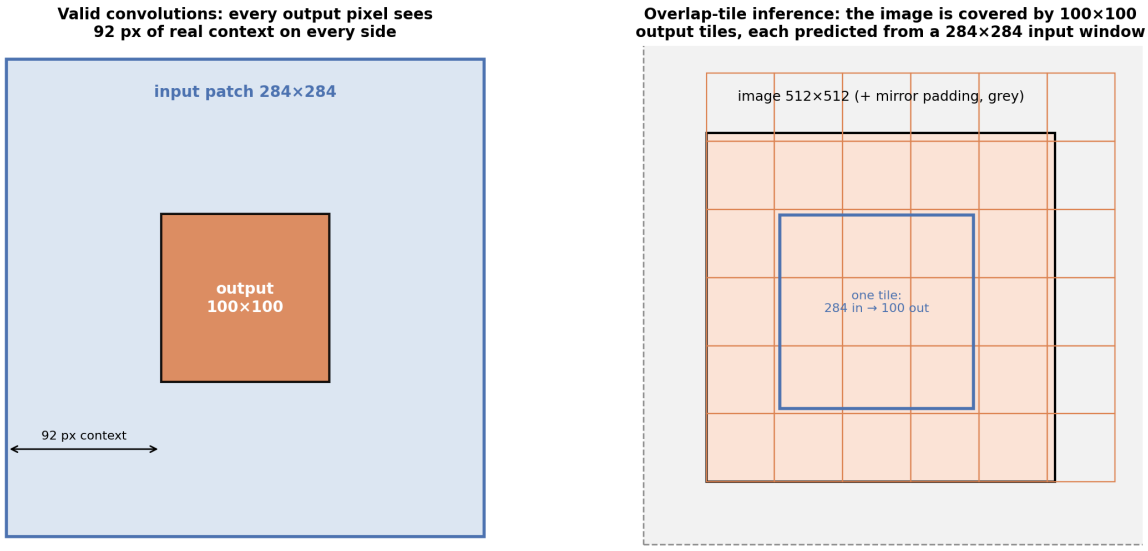


Figure 3. Left: a 284×284 input patch predicts the central 100×100 region. Right: overlap-tile inference covers the whole image with 100×100 output tiles.

6.3 Sampling and augmentation

- **FOV-centred sampling**: patch centres are chosen inside the retina, never in the black background.

- **Fresh patches every epoch**: 6 new random patches per image each epoch (2,880 per epoch), so over 30 epochs the network sees far more of each retina.
- **Augmentation** (creating new realistic variations of training patches): random horizontal/vertical flips and 90° rotations (vessels can run in any direction) and **elastic deformation** (smoothly warping the patch, as in the U-Net paper, which mimics natural variation in vessel shapes). Augmentation is applied only during training and reduces overfitting.
- All random steps use fixed seeds, so every run is exactly reproducible.

6.4 Predicting a full image (overlap-tile strategy)

To segment a whole test image, it is mirror-padded by 92 pixels (the border is reflected, so the edge regions still have realistic context), then covered by a grid of 100×100 output tiles, each predicted from its 284×284 input window. The tiles are placed side by side into a full probability map, pixels outside the FOV are set to 0, and finally each pixel with probability ≥ threshold is marked as vessel.

7. The network architectures

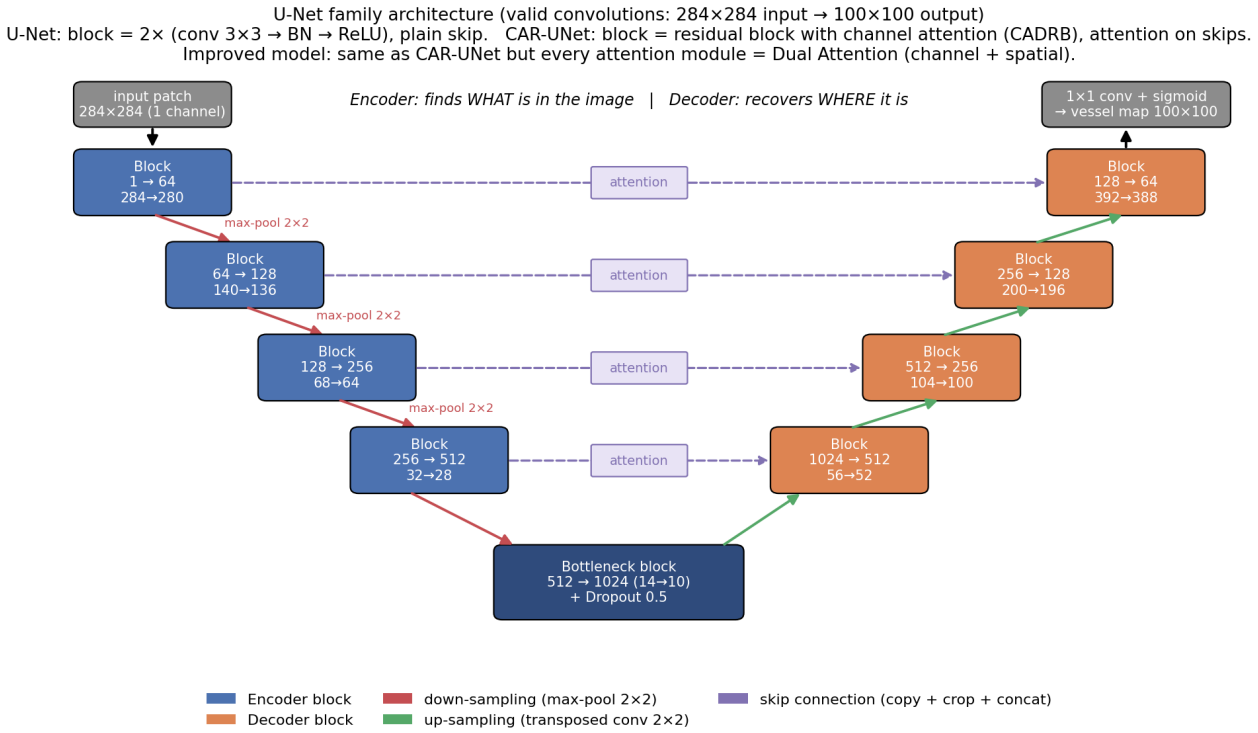


Figure 4. The U-Net family architecture used in this project, with feature-map channels and sizes for a 284×284 input patch.

7.1 U-Net (Review 1 and reference model)

- **Encoder (contracting path)**: 4 blocks of two 3×3 convolutions (+BN+ReLU), each followed by 2×2 max-pooling. Channels double each time (64→128→256→512) while the image shrinks. The encoder learns what is in the image.
- **Bottleneck**: the deepest block (1024 channels) with dropout 0.5.
- **Decoder (expanding path)**: 4 blocks that up-sample with 2×2 transposed convolutions and halve the channels. The decoder recovers where things are, at full resolution.
- **Skip connections**: each encoder output is copied (centre-cropped) and joined to the decoder at the same level. Fine details lost by pooling come back this way — essential for thin vessels.
- Final 1×1 convolution + sigmoid → probability of vessel for every pixel. About **31.0 million** parameters.

7.2 CAR-UNet (the paper model)

- **Residual blocks (CADRB)**: each block adds its input back to its output (a shortcut). The block only has to learn the correction, which makes deep networks easier to train and keeps information flowing.
- **Channel attention (MECA)**: after the convolutions, each feature channel gets a learned weight between 0 and 1 (computed from the channel's average with a small 1D convolution). Useful channels are amplified, useless ones suppressed.
- **Attention on skip connections**: encoder features are re-weighted before they are joined to the decoder, instead of copied unchanged.
- About **32.4 million** parameters (≈1.4 M more than U-Net).

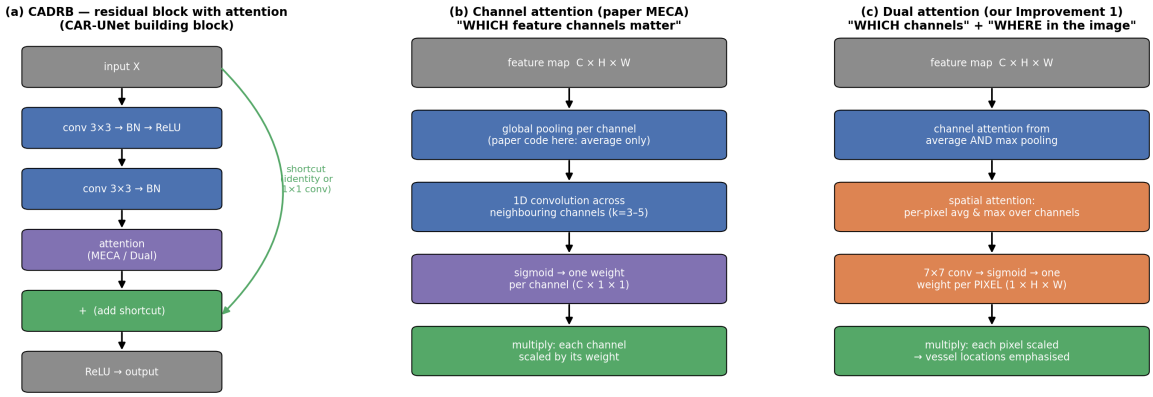


Figure 5. (a) The residual block with attention. (b) Channel attention of the paper. (c) Our Dual Attention (Improvement 1).

8. Loss functions and training

8.1 The class-imbalance problem

Only about **8–9% of the pixels** in a fundus image are vessel. A lazy model that predicts 'background' everywhere would already be 91% accurate while finding no vessels at all. The loss function must therefore not let the majority class dominate.

Loss	Idea	Strength / weakness
Binary cross-entropy (BCE)	Penalises each pixel's probability error separately	Stable, smooth gradients — but dominated by the many background pixels
Dice loss	1 − overlap between predicted and true vessel regions	Ignores the background imbalance by design — but alone can be unstable
0.5·BCE + 0.5·Dice (used)	Combination of both	**Stable training and good overlap; standard choice for vessel segmentation**
Weighted BCE / Focal / Tversky	Re-weight rare pixels or hard pixels	Valid alternatives; need extra tuning; BCE+Dice works well here
+ clDice (Improvement 2)	Overlap of vessel skeletons	Rewards connected vessels (see Section 12)

8.2 Training settings (identical for every model)

Setting	Value	Why
Optimiser	Adam, learning rate 0.001, weight decay 0.00001	Reliable default for U-Net-type models; weight decay discourages overly large weights
Learning-rate schedule	Cosine annealing over 30 epochs	Big steps early, tiny steps at the end for fine convergence
Batch size / epochs	8 patches, 30 epochs	Fits in 6 GB GPU memory; validation Dice levels off by epoch ~25
Model selection	Epoch with the best mean Dice on the 120 full validation images	Avoids keeping an over-trained last epoch; never uses test images
Repeats	3 random seeds per model	Shows the result is not luck; gives mean ± standard deviation
Hardware	NVIDIA RTX 4050 laptop GPU (6 GB)	≈71 min (U-Net) to ≈195 min (1024 model) per training run

> **Why the same settings for every model?**
> If one model trains longer or with more data than another, any difference could come from the training, not from the model. In the first version of the project U-Net and CAR-UNet had different settings (5 vs 15 epochs). Using one script and one configuration for all models makes the comparison fair.

9. How results are measured

9.1 The confusion matrix

For every pixel inside the FOV we compare the prediction with the expert tracing: **TP** (vessel, correctly found), **FP** (background wrongly marked as vessel), **FN** (vessel missed), **TN** (background correctly left out). All metrics are built from these four counts.

Metric	Formula	Meaning	Why we report it
Accuracy	(TP+TN) / all	Fraction of pixels classified correctly	Reported by the paper — but misleading with 91% background
Sensitivity (recall)	TP / (TP+FN)	Fraction of vessel pixels found	Missing vessels is the main clinical risk
Specificity	TN / (TN+FP)	Fraction of background left out	Reported by the paper
Precision	TP / (TP+FP)	Fraction of detected pixels that are real vessels	Measures false alarms
Dice (F1)	2TP / (2TP+FP+FN)	Overlap of prediction and truth (0–1)	**Main metric**: balances found vessels and false alarms
IoU (Jaccard)	TP / (TP+FP+FN)	Stricter overlap measure	Common in segmentation papers
AUC-ROC / AUC-PR	Area under curve over all thresholds	Quality of the probabilities, independent of the threshold	Paper uses AUC-ROC; AUC-PR suits imbalanced data
clDice	Overlap of skeletons (centrelines)	How well vessel connectivity is preserved	Thin vessels count as much as thick ones
Thin-vessel recall	TP on thin vessels / thin vessels	Fraction of thin vessels (radius ≤ 1.5 px) found	Directly measures our main goal

9.2 Making the comparison trustworthy

- Metrics are computed **per image** and averaged over all **200 test images** (50 per disease), and also reported **per disease**.
- Each model is trained with **3 seeds**; we report mean ± standard deviation.
- **Paired Wilcoxon signed-rank test**: for each test image we compare both models on the same image and test whether the improved model is consistently better. It is paired (same images) and non-parametric (it does not assume the differences are normally distributed, which Dice differences are not). A **p-value** below 0.05 means a difference this consistent would be very unlikely by chance.
- Every saved evaluation is re-checked: running a saved model again must reproduce its stored scores (the built-in sanity check).

10. Fixing the evaluation before improving the model

Before improving anything we audited the first version of the FIVES experiment and found problems that made its numbers unreliable. All were fixed, and U-Net and CAR-UNet were retrained (3 seeds each) under the corrected protocol.

Problem in the first version	Why it mattered	Fix
Only 50 of 200 test images were scored — 45 of them Glaucoma	Results were not a fair multi-disease test	All 200 test images (50 per disease)
Validation set = 68 Normal + 52 AMD only	Best model chosen without DR or Glaucoma	Disease-stratified split (30 each)
Different training settings for the two models	Differences could come from training, not the model	One script, identical settings
One run, no seed	Could not tell a real gain from luck	3 seeds, mean ± std, significance tests
Same patches reused every epoch	Less training variety	Fresh random patches each epoch
Elastic augmentation ignored the seed	Runs not exactly reproducible	Seeded random generator

Result after the fixes (200 images, 3 seeds)	U-Net	CAR-UNet (paper)
Dice	0.8344 ± 0.0034	0.8382 ± 0.0031
Sensitivity	0.8247	0.8310
clDice	0.8593	0.8626
CAR-UNet better on		154 / 200 images ($p = 2\times10^{-14}$)

The first version had reported a much larger gain (Dice 0.774 → 0.792). Under the fair protocol, CAR-UNet is still better, but only slightly — and its error maps showed that almost all remaining errors are **missed thin vessels**. This set the direction for the improvements.

11. Limitations of the paper

#	Weakness of CAR-UNet	Evidence	Fixed by
W1	Channel attention only: it learns which features matter but has no information about where in the image the vessels are	Paper §2.2; missed thin vessels in our error maps	Improvement 1
W2	Pixel-wise BCE loss: vessels are ~8% of pixels and nothing rewards keeping vessels connected — a broken thin vessel costs only a few pixels	Paper §3.2; broken capillaries at vessel tips	Improvement 2
W3	Designed for small low-resolution images; on FIVES the 2048×2048 images must be shrunk, which erases the finest capillaries	Missed vessels are almost all thin	Improvement 3
W4	Fixed 0.5 decision threshold and a single prediction	Paper §3.3	Improvement 4
W5	Weak evaluation: 20 training images, only Accuracy/Sensitivity/Specificity/AUC, no Dice, no connectivity, no per-disease results, one run	Paper Tables 1–3	Our protocol (Sections 9–10)

Honest note on our paper implementation
Our paper model follows the CAR-UNet design (residual blocks, channel attention in every block and on every skip) but, like Review 1, uses the U-Net valid-convolution geometry, 64 base filters and Dropout. Its channel attention uses average pooling; the paper's MECA uses average and max pooling. Improvement 1 includes the paper's average+max pooling, so the improved model is at least as close to the paper as the base model.

12. The four improvements (what, how, why)

Improvement 1 — Dual attention (fixes W1)

Idea. Add a second attention step that tells the network where vessels are. **How.** Every attention module becomes: (1) channel attention computed from the **average and the maximum** of each channel (max-pooling captures the strongest, most distinctive responses — the paper itself motivates this), followed by (2) **spatial attention** (from CBAM, Woo et al. 2018): for every pixel, the average and maximum over all channels are combined by a 7×7 convolution into **one weight per pixel**. Pixels in vessel-like structures are treated differently from background. **Cost:** only 1,274 extra parameters.

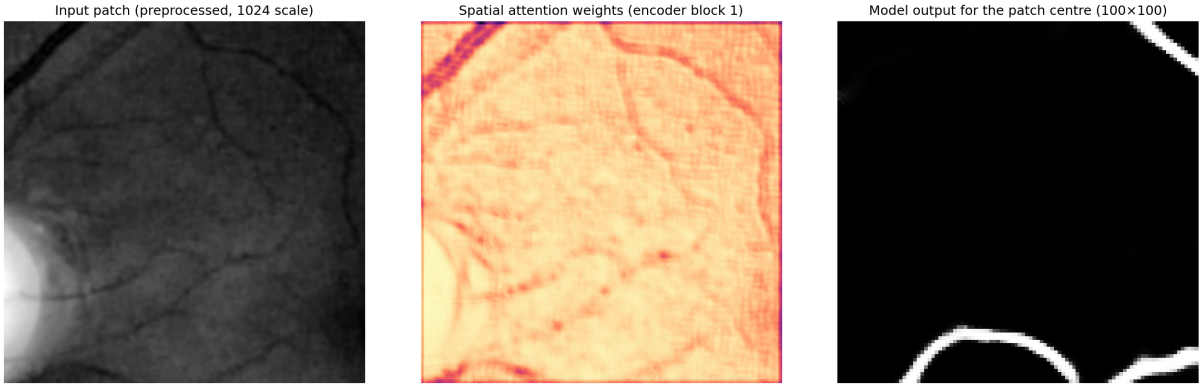


Figure 6. The spatial attention map of the improved model (middle) traces the vessel tree, including thin vessels — the network has learned where vessels are.

> **Why this and not something else?**
> Alternatives: **attention gates** (Attention U-Net) only act on skip connections and add more parameters; **self-attention / transformers** need much more memory and data. Spatial attention is tiny, plugs into the existing CAR-UNet blocks without changing anything else, and directly fixes the missing 'where' information.

Result (3 seeds): Dice +0.0026 (better on 167/200 images, $p = 2\times10^{-19}$), precision +0.006 (fewer false vessels) and the most stable results across seeds.

Improvement 2 — clDice loss (fixes W2)

Idea. Reward the network for keeping vessels connected. **How.** clDice (Shit et al., CVPR 2021) reduces both the prediction and the expert tracing to their **skeletons** (1-pixel-wide centrelines) using a differentiable 'soft skeleton' (repeated soft erosion and dilation), then measures how much of each skeleton lies inside the other region. A broken thin vessel removes a whole piece of skeleton, so the loss rises sharply. The training loss becomes 0.5·BCE + 0.5·(0.5·Dice + 0.5·clDice).

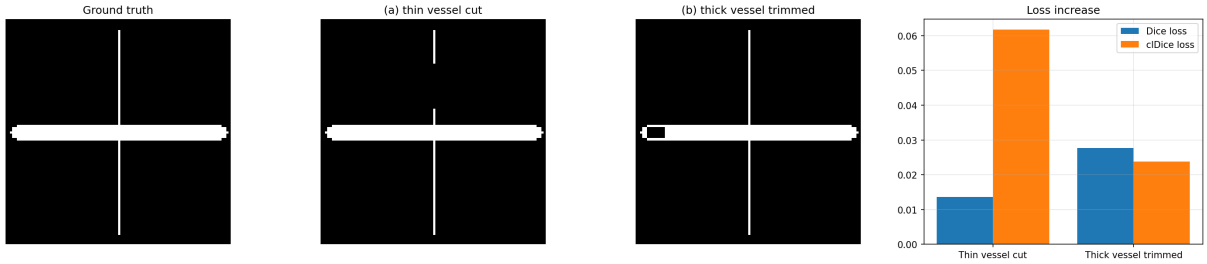


Figure 7. Synthetic test: cutting a thin vessel (20 pixels) raises the clDice loss 4.5× more than the Dice loss; trimming a thick vessel costs less under clDice.

> **Why this and not something else?**
> Alternatives: **topological losses** based on persistent homology are slow and complex; **post-processing** (joining broken vessels afterwards) needs hand-tuned rules. clDice is fast, differentiable and designed exactly for tubular structures such as vessels.

Result (3 seeds): connectivity (clDice) +0.0088 (better on 191/200 images, $p = 2\times10^{-32}$) and sensitivity +0.0124. Dice stays level because the model now also marks some extra pixels at the default threshold — a trade-off that Improvements 3 and 4 then resolve.

Improvement 3 — 1024×1024 working resolution (fixes W3)

Idea. Do not throw away the detail of the original images. **How.** The improved model is trained and run on 1024×1024 copies (from the 2048×2048 originals) instead of 512×512. A thin capillary that is a fraction of a pixel wide at 512 is about twice as wide at 1024 and survives. For a fair comparison its prediction is **shrunk back and scored on exactly the same 512×512 ground truth** as the paper model; we additionally score both against the 1024×1024 ground truth. The clDice skeleton uses 20 iterations at 1024 because vessels are twice as wide in pixels.



Figure 8. The same region at 512 and 1024: at 512 thin vessels in the ground truth are broken into pieces; at 1024 they are continuous.

> **Why this and not something else?**
> Full 2048×2048 would keep even more detail but needs about 4× more time and memory than 1024 — too much for the laptop GPU. Super-resolution networks add a whole second model. 1024 is the best trade-off between detail and cost (training ≈1.6× longer).

Result (3 seeds): Dice +0.0040 over the same model at 512 (better on 150/200 images, $p = 2\times10^{-11}$), precision +0.020, and against the 1024×1024 ground truth Dice 0.8796 vs 0.8599 (same model at 512) and 0.8572 (paper model).

Improvement 4 — threshold chosen on the validation set (fixes W4)

Idea. The network outputs probabilities; 0.5 is an arbitrary cut-off. **How.** For each trained model we test thresholds from 0.20 to 0.80 on the **120 validation images only** and keep the one with the best Dice. We also tested **test-time augmentation (TTA)**: averaging the predictions on 4 flipped copies of each image; the validation set decides whether to use it.

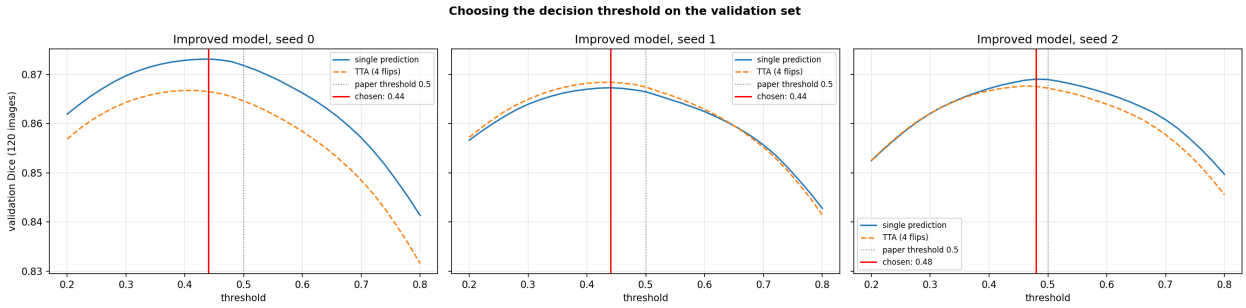


Figure 9. Validation Dice versus threshold for the three improved models; the red line marks the threshold chosen on validation.

Result: the 1024 models are slightly cautious, so validation chooses 0.44–0.48 → sensitivity +0.014 and Dice +0.0015 (better on 147/200 images). **Honest negative result:** TTA lowered Dice for most models. A diagnostic showed the models segment flipped images up to ~2% worse despite flip augmentation, so averaging adds weaker predictions. It is therefore used only where validation shows a gain (1 of 3 final models).

13. Final results

13.1 Paper model vs improved model (200 test images, mean ± std over 3 seeds)

Metric	Paper model	Improved model	Change	Better on	p-value	Winner
Accuracy	0.9728	0.9737	+0.09	130/200	1.8e-08	Improved
Sensitivity	0.8310	0.8465	+1.55	159/200	7.4e-19	Improved
Specificity	0.9855	0.9846	−0.09	51/200	6.3e-15	Paper
Precision	0.8557	0.8516	−0.40	68/200	9.6e-08	Paper
Dice / F1	0.8382 ± 0.0031	**0.8436 ± 0.0027**	**+0.54**	140/200	4.8e-10	Improved
IoU	0.7295	0.7379	+0.83	140/200	2.0e-10	Improved
AUC-ROC	0.9860	0.9878	+0.17	168/200	1.4e-21	Improved
AUC-PR	0.9202	0.9260	+0.58	163/200	9.4e-18	Improved
clDice	0.8626 ± 0.0057	**0.8751 ± 0.0005**	**+1.25**	175/200	5.3e-25	Improved
Thin-vessel recall (seed 0)	0.7806	0.7883	+0.77	139/200	2.7e-07	Improved
Dice vs 1024 ground truth	0.8572	0.8796	**+2.24**	—	—	Improved

Changes are in percentage points (0.01 = 1 point). The improved model wins on **9 of 11** metrics. The two losses (precision −0.40, specificity −0.09) are the expected price of finding more vessels: a model that detects more thin vessels also marks a few more background pixels.

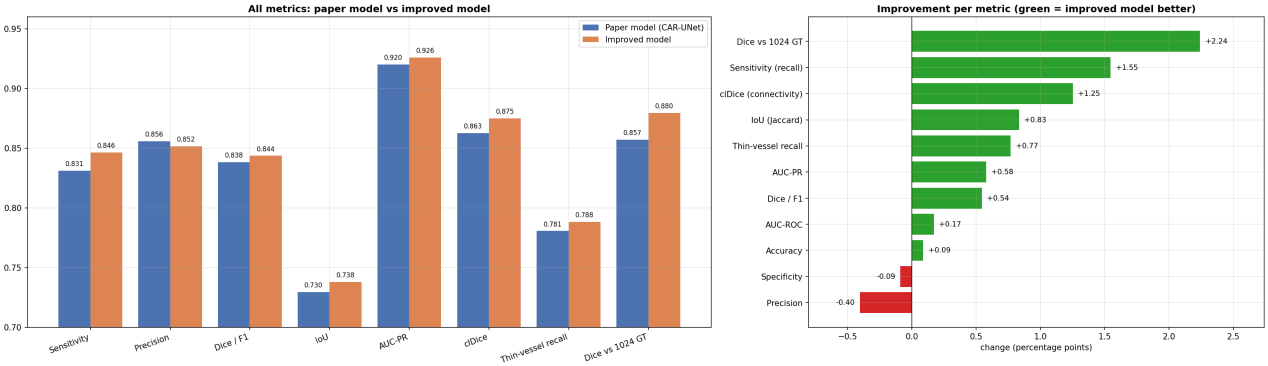


Figure 10. Left: all metrics side by side. Right: change per metric (green = improved model better).

13.2 Ablation: what each improvement adds

Step (3 seeds each)	Dice	Sensitivity	Precision	clDice
Paper model (CAR-UNet)	0.8382	0.8310	0.8557	0.8626
+ Dual attention	0.8408	0.8308	0.8620	0.8631
+ clDice loss	0.8382	0.8432	0.8430	0.8718
+ 1024 resolution	0.8421	0.8326	0.8632	0.8742
+ tuned threshold = improved model	0.8436	0.8465	0.8516	0.8751

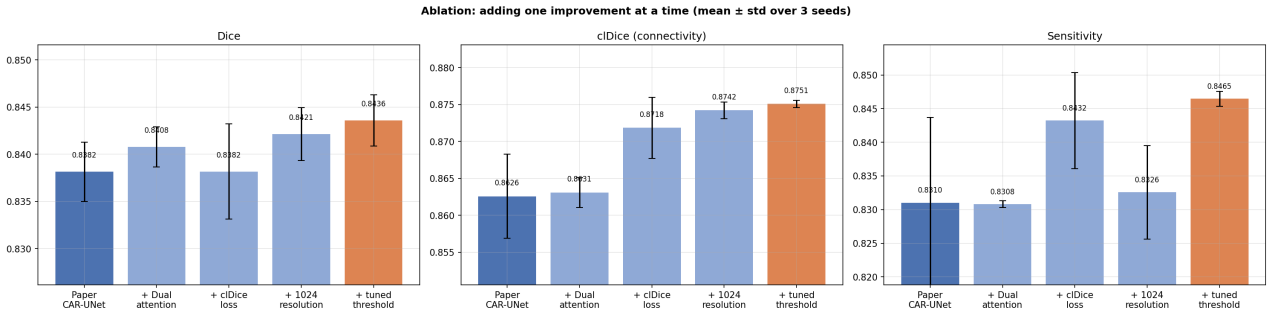


Figure 11. Ablation bar charts (mean ± std over 3 seeds).

13.3 Every test image, both models

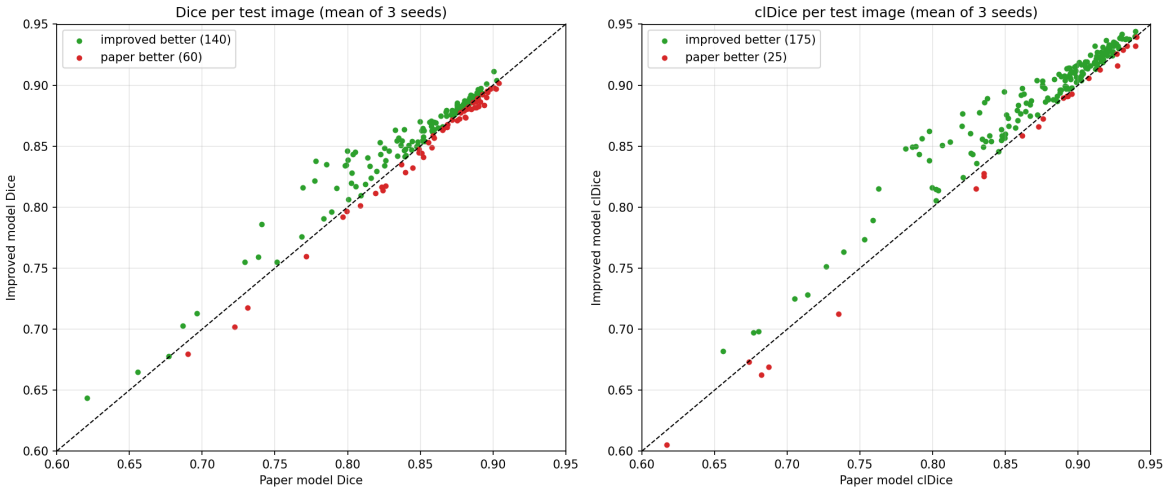


Figure 12. Each point is a test image. Points above the diagonal: the improved model is better (Dice: 140/200, clDice: 175/200).

13.4 Per disease

Disease	Paper Dice	Improved Dice	Change	Paper clDice	Improved clDice	Change
AMD	0.8643	0.8671	+0.28	0.8884	0.8978	+0.94
DR	0.8562	0.8585	+0.22	0.8807	0.8895	+0.88
Glaucoma	0.7958	0.7947	−0.11	0.8178	0.8227	+0.49
Normal	0.8363	0.8541	+1.78	0.8635	0.8904	+2.69

Connectivity improves in every disease group. Dice improves most on Normal images; Glaucoma is essentially unchanged and remains the hardest group (its images often have low contrast and fewer visible vessels).

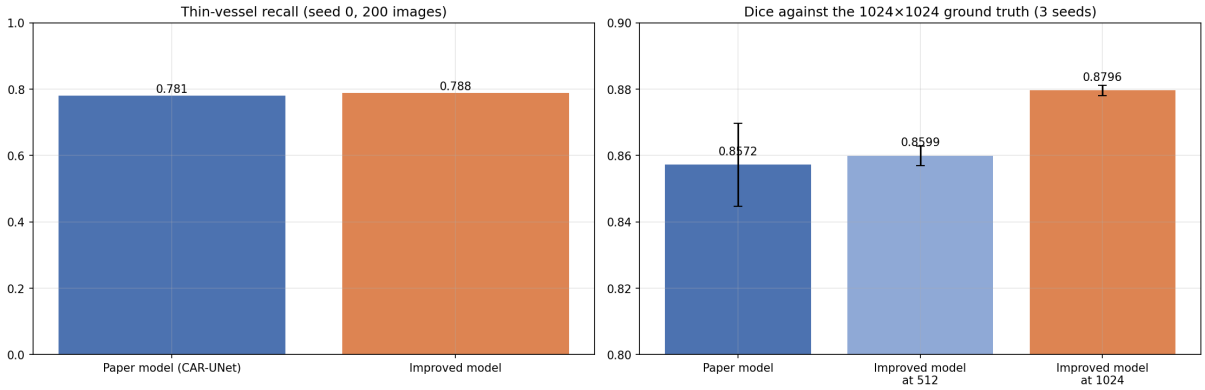


Figure 13. Thin vessels: thin-vessel recall (left) and Dice against the 1024×1024 ground truth (right).

13.5 What the outputs look like

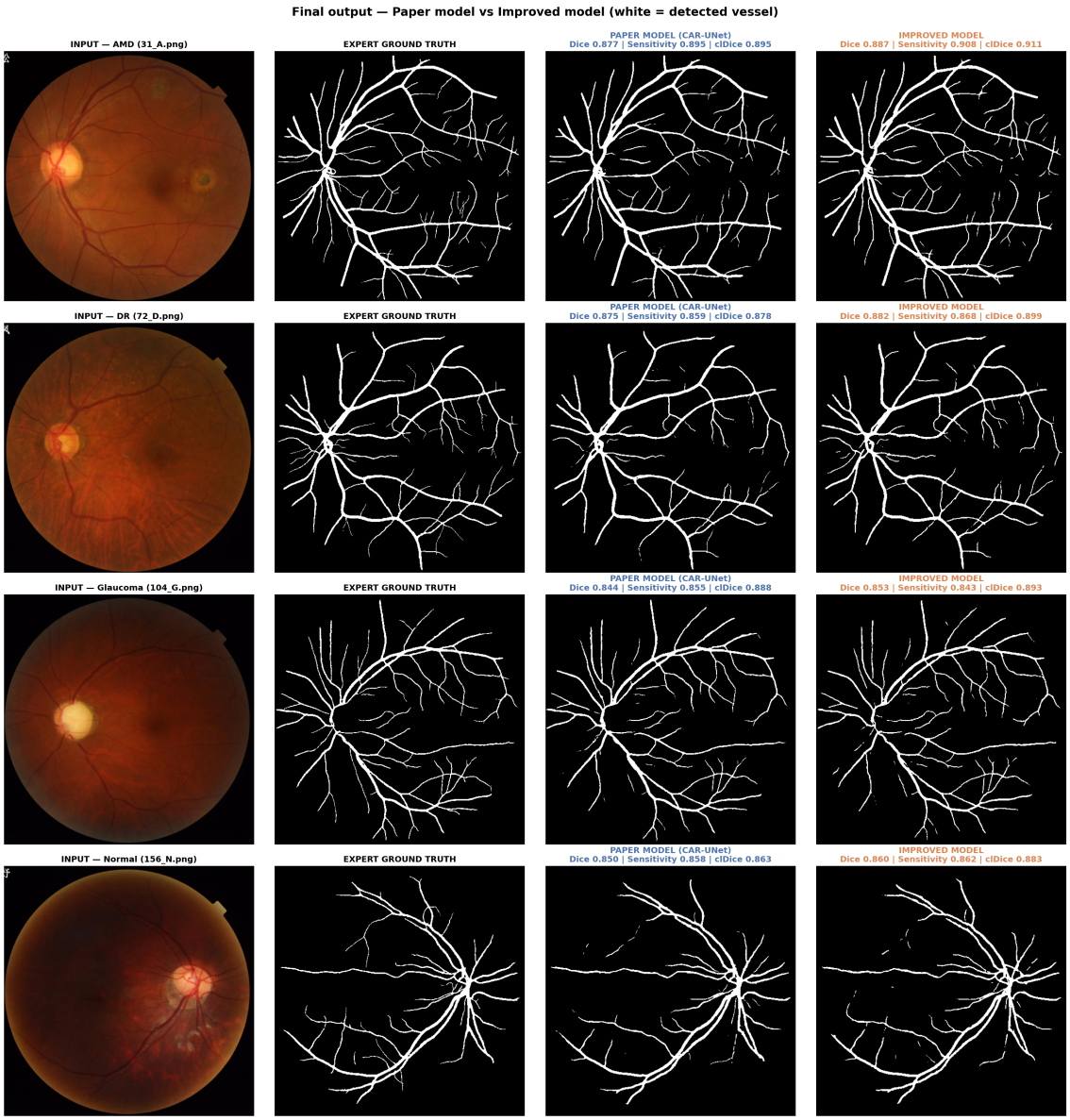


Figure 14. One typical test image per disease (median paper-model Dice — not hand-picked): input, expert tracing, paper model, improved model.

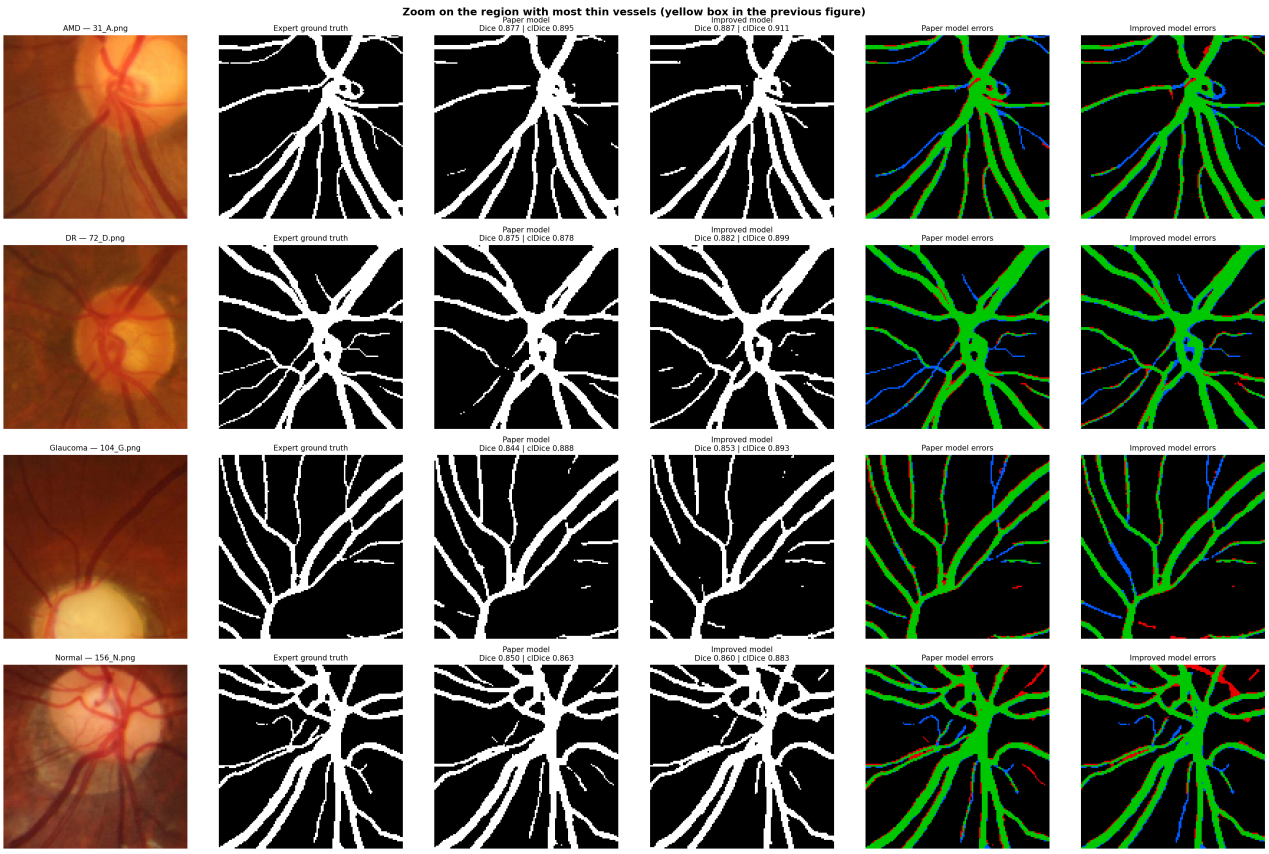


Figure 15. Zoom on the region with the most thin vessels. Green = correct vessel, red = false vessel, blue = missed vessel. The improved model has fewer blue (missed) thin vessels.

14. Honest limitations and future work

- **Small absolute Dice gain** (+0.54 points): Dice is dominated by thick vessels, which both models already segment well; the gains are larger on thin vessels and connectivity, which matter clinically.
- **Glaucoma is not improved**, and two very low-quality test images (122_G, 123_G) fail for every model (Dice below 0.1) because almost no vessels are visible.
- **Precision drops slightly** (−0.40 points) because more vessels are detected.
- **Cost**: the 1024 model needs about 1.6× the training time and about 4× more tiles at prediction time.
- **Future work**: extra weight for hard/Glaucoma images, using the FIVES image-quality labels, full 2048×2048 training on a larger GPU, and a lighter model (the original paper uses 16 base filters instead of 64).

15. Questions your sir may ask — with answers

Q: What is the aim of the project?

To automatically segment retinal blood vessels in fundus images, replicate the CAR-UNet paper on the FIVES dataset, find its weaknesses and improve it — especially for thin vessels and vessel connectivity.

Q: Why segmentation and not classification?

Classification gives one label per image (e.g. 'diabetic'); segmentation gives a label per pixel, producing a vessel map from which doctors can measure vessel width, branching and breaks.

Q: Why U-Net?

It was designed for biomedical segmentation with few training images: the encoder captures context, the decoder recovers exact positions, and skip connections bring back fine details. It is the standard base for vessel segmentation.

Q: What do skip connections do?

They copy high-resolution encoder features directly to the decoder, so details lost by pooling (like 1–2 pixel wide vessels) can be recovered.

Q: What is a residual block and why use it?

A block that adds its input to its output. It only has to learn a correction, gradients flow easily through the shortcut, and deeper networks train reliably.

Q: What is channel attention?

A small module that gives every feature channel a weight between 0 and 1, so useful features are amplified and useless ones suppressed. It answers 'which features matter'.

Q: What does your dual attention add?

Spatial attention: one weight per pixel, computed from the average and maximum over channels with a 7×7 convolution. It answers 'where in the image' — the attention map visibly traces the vessel tree.

Q: Why the green channel?

Haemoglobin absorbs green light strongly, so vessels appear darkest and with the highest contrast in the green channel; red is over-saturated and blue is noisy.

Q: Why CLAHE and not normal histogram equalisation?

Fundus images are unevenly lit. CLAHE enhances contrast locally in small tiles and clips the enhancement to avoid amplifying noise, so thin vessels become visible everywhere without over-brightening the optic disc.

Q: Why a bilateral filter?

It removes noise but preserves edges, because it only averages pixels with similar intensity. A Gaussian blur would blur thin vessels away.

Q: Why patches instead of whole images?

Patches fit in GPU memory, give thousands of varied training samples, and vessels are a local pattern. Whole images are predicted with the overlap-tile strategy.

Q: Why 284×284 input and 100×100 output?

We use valid convolutions (no padding) as in the original U-Net. The network shrinks the patch by 184 pixels, so every output pixel has 92 pixels of real context on each side and there are no border artefacts.

Q: Why data augmentation?

To create realistic variations (flips, rotations, elastic warps) so the network learns general vessel patterns instead of memorising the training images.

Q: How did you avoid data leakage?

The split is per image (never per patch), stratified by disease with a fixed seed, and the test images are never used for training or for any choice — the best epoch and the threshold are chosen on validation images.

Q: Why BCE + Dice loss?

Vessels are only ~8% of pixels. BCE gives stable pixel-wise gradients; Dice measures overlap and is not dominated by the background. Together they train stably and handle the imbalance.

Q: What is clDice and why did you use it?

clDice compares the skeletons (centrelines) of prediction and ground truth. A broken thin vessel loses a whole piece of skeleton, so the loss rises sharply — it directly rewards connected vessels, which BCE and Dice do not.

Q: Why is accuracy not your main metric?

With ~91% background pixels, a model that finds no vessels is still 91% accurate. Dice, sensitivity and clDice focus on the vessels themselves.

Q: Why compute metrics only inside the FOV?

The black area outside the retinal circle is trivially easy background; including it inflates accuracy and specificity without saying anything about vessel quality.

Q: Why 3 seeds?

Training involves randomness (initial weights, patches, augmentation). Three runs show whether a difference is consistent or just luck, and give a standard deviation.

Q: What does the Wilcoxon test tell you?

Whether the improved model is consistently better on the same test images. It is paired and does not assume normally distributed differences. All our main improvements have p far below 0.05.

Q: Why 1024 and not 2048?

2048 would need about 4× more time and memory than 1024, which does not fit a 6 GB laptop GPU in reasonable time. 1024 already recovers most of the lost thin vessels.

Q: How can you compare a 1024 model with a 512 model fairly?

The 1024 prediction is shrunk to 512 and scored on exactly the same 512 ground truth as the paper model; we also score both against the 1024 ground truth.

Q: Why did TTA not help?

Our diagnostic showed the models segment flipped images up to ~2% worse, so averaging flipped predictions adds weaker predictions. We report this honestly and let the validation set decide when to use it.

Q: Why is precision slightly lower for the improved model?

It finds more vessels (sensitivity +1.55 points), and some extra detections are false. Dice, which balances both, still improves.

Q: The Dice gain is only 0.54 points — is that meaningful?

Yes: it is consistent (better in all 3 seeds, $p = 4.8\times10^{-10}$), and Dice is dominated by thick vessels. On what the improvements target the gains are larger: connectivity +1.25, sensitivity +1.55, Dice against the 1024 ground truth +2.24.

Q: How is your work different from the paper?

Same base architecture, but we add spatial attention, a connectivity loss, higher resolution and a validation-tuned threshold, evaluate on a much larger multi-disease dataset, and add Dice, clDice, per-disease results, 3 seeds and significance tests.

Q: What would you do next?

Focus on Glaucoma and low-quality images (e.g. using FIVES quality labels), try full-resolution training on a bigger GPU, and build a lighter, faster model.

16. Glossary

Term	Simple meaning
Fundus image	Photograph of the back of the eye (retina)
Segmentation	Labelling every pixel (here: vessel or background)
Ground truth	The expert's manual vessel tracing used as the correct answer
FOV	Field of view: the circular retina region in the photo
Feature map / channel	Output of a convolution filter; a layer with 64 filters has 64 channels
Encoder / decoder	Down-sampling path (what) / up-sampling path (where)
Skip connection	Shortcut copying encoder features to the decoder
Residual block	Block that adds its input to its output
Attention	Learned weights that emphasise important channels (channel attention) or locations (spatial attention)
Loss	Number measuring how wrong the prediction is; training minimises it
Epoch / batch	One pass over the data / examples processed together
Overfitting	Memorising training data instead of learning general rules
Augmentation	Making new training examples by flipping, rotating, warping
Seed	Starting value of the random generator; same seed → same run
Ablation	Adding (or removing) one component at a time to measure its effect
Skeleton	1-pixel-wide centreline of a vessel
p-value	Probability of seeing such a consistent difference by chance; below 0.05 = significant
TTA	Test-time augmentation: averaging predictions on flipped copies
Threshold	Probability above which a pixel is called vessel

17. Where to find everything in the project

What	File
Paper vs improved comparison (all outputs, plots, code)	Final_Comparison_Paper_vs_Improved.ipynb
Stage 2 notebook (U-Net vs CAR-UNet + improvements summary)	Review2_Complete_CAR_UNet_Pipeline.ipynb
Review 1 notebook (U-Net on DRIVE)	main_notebook.ipynb
Improvement report / slide content / overview	IMPROVEMENTS.md, SLIDES_OUTLINE.md, README.md
Models (dual attention)	src/car_unet.py
Losses (BCE + Dice, clDice)	src/losses.py
Preprocessing / data loading / metrics	src/preprocessing_fives.py, src/data_loader_fives.py, src/metrics.py
Training (512 and 1024) / threshold + TTA evaluation	scripts/train_fives.py, scripts/run_baselines.py, scripts/evaluate_post.py
Result tables and tests / figures	results/RESULTS_TABLE.md, results/POST_RESULTS.md / outputs/*.png